ROTDIF-Web: A GenApp Generated Science Gateway for Comprehensive Analysis of Biomolecular NMR Relaxation Data

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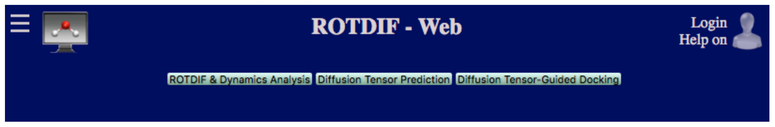
INTRODUCTION: Structure and dynamics play critical role in biomolecular interactions, and accurate characterization of the overall rotational diffusion and local motions from NMR spin-relaxation data can help refine biomolecular structures and serve as restraints in protein docking. Using the GenApp technology for scientific gateways [1] here we introduce ROTDIF-Web, a science gateway for such analysis, that integrates previously developed standalone downloadable programs, written in Matlab or Java [2,3], and provides advanced computational functionalities, streamlines data input, storage, and output, and enables interactive 2D and 3D plotting and visualization of results.

OVERVIEW: ROTDIF-Web is a versatile software package that enables researchers to perform accurate and comprehensive analysis of NMR spin-relaxation data. It has three main features/modules:

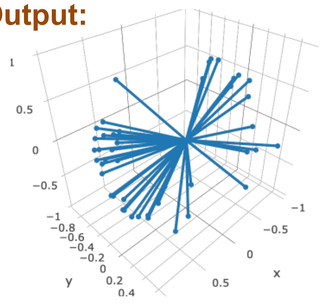
- ROTDIF & DYNAMICS: Determine the overall rotational diffusion tensor and characterize local dynamics in proteins and nucleic acids [2,3].
- ELM predictor: *Ab initio* prediction of rotational diffusion tensors directly from macromolecule structures [4].
- ELMDOCK: Build macromolecular complexes using rigid-body docking guided by experimental rotational diffusion tensors [5,6].



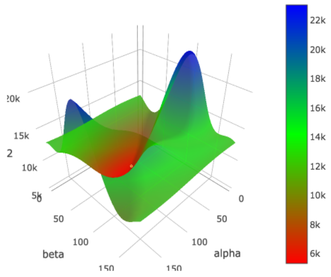
Module 1: ROTDIF & Dynamics Analysis

Input: macromolecular structures (e.g. from Protein Data Bank) and NMR spin-relaxation data

Output:

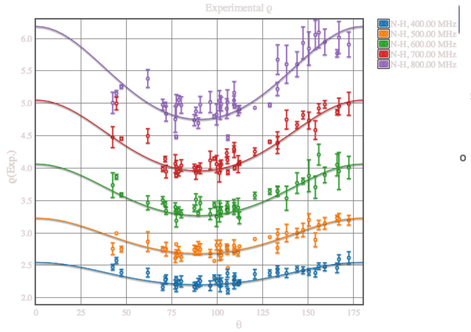


Plot of bond vector orientations

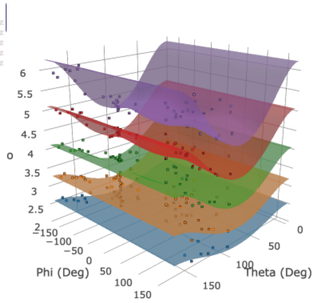


Chi-square plot

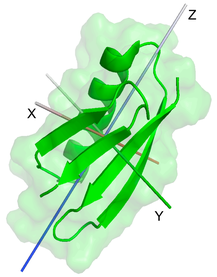
Axially Symmetric Model Fit



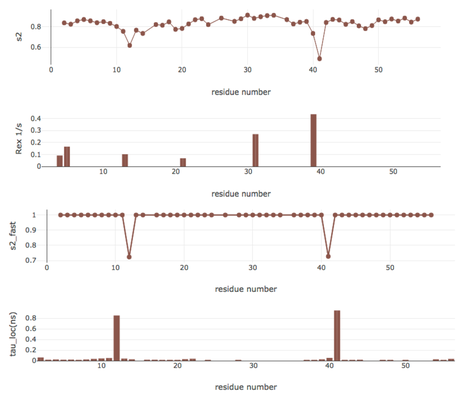
Fully Anisotropic Model Fit



Diffusion tensor axes overlaid onto the protein structure (PDB ID: 1P7F)



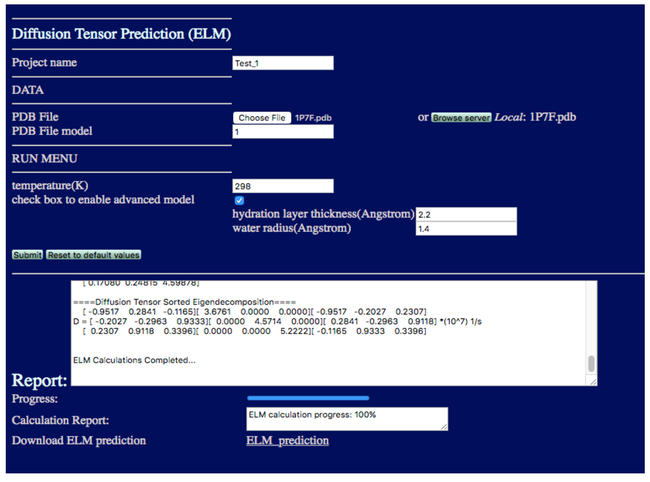
The Results of Dynamics Analysis



Module 2: Diffusion Tensor Prediction (ELM)

Input: macromolecule structures (e.g. PDB files)

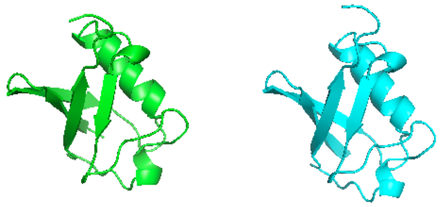
Output: predicted diffusion tensor



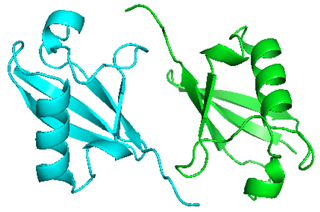
Module 3: Diffusion Tensor-Guided Docking

Input: macromolecule structures (e.g. PDB files) and NMR relaxation data

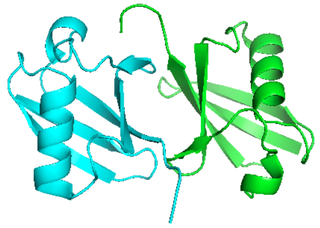
Output: Docked protein complexes



Ubiquitin monomers before docking (PDB ID: 1D3Z)



ELMDOCK-produced diffusion-guided docked structure (T=301K)



Crystal structure of ubiquitin dimer from Protein Data Bank(PDB ID:1AAR)

References:

[1] A. Savelyev and E. H. Brookes. Fut. Gen. Comp. Sys. (2017)
[2] O. Walker et al, J Magn Reson *168*, 336 (2004).
[3] K. Berlin et al., J Biomol NMR *57*, 333 (2013).
[4] Ya. Ryabov et al., J Am Chem Soc 128, 15432 (2006).
[5] Ya. Ryabov et al., J Am Chem Soc 129, 7894 (2007).
[6] K. Berlin et al., Proteins *79*, 2268 (2011).

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